

Meritve $\{z_i\}_i \dots$ vzorce

$$z_i \sim \mathcal{N}(\mu, \sigma^2)$$

$$\sigma_{ij} = 0$$

a) ocenjevanje parametrov normalne porazd.

• cenilka za pričakovano vrednost μ :

$$\bar{z} = k \sum z_i$$

$$\langle \bar{z} \rangle = k \sum \langle z_i \rangle = k \sum \mu = k N \mu \stackrel{\text{da}}{=} \mu \rightarrow k = \frac{1}{N}$$

• cenilka za varianco σ^2

1) poznano μ

$$\hat{\sigma}^2 = k \sum (z_i - \mu)^2$$

$$\langle \hat{\sigma}^2 \rangle = k \sum \langle (z_i - \mu)^2 \rangle = k N \sigma^2 = \sigma^2 \rightarrow k = \frac{1}{N}$$

2) Ne poznano μ

$$s^2 = k \sum (z_i - \bar{z})^2$$

$$\langle s^2 \rangle = \sigma^2$$

$$\langle s^2 \rangle = k \sum \langle (z_i - 2z_i \bar{z} + \bar{z}^2) \rangle = k \sum \langle z_i^2 - 2z_i \bar{z} + \bar{z}^2 \rangle = k \sum \langle z_i^2 - 2\bar{z} \sum z_i + N \bar{z}^2 \rangle =$$

$$= k \sum \langle z_i^2 - N \bar{z}^2 \rangle = k \sum \langle (z_i^2 - \bar{z}^2) \rangle = k \sum \langle z_i^2 - \langle \bar{z}^2 \rangle \rangle$$

$$\sigma^2 = \langle (z_i - \mu)^2 \rangle \rightarrow \langle z_i^2 \rangle = \sigma^2 + \mu^2 + 2\mu^2 = \sigma^2 + \mu^2$$

$$\text{Var}(\bar{z}) = \langle (\bar{z} - \mu)^2 \rangle$$

$$\frac{\sigma^2}{N} = \langle \bar{z}^2 \rangle + \mu^2 - 2\mu^2 \rightarrow \langle \bar{z}^2 \rangle = \mu^2 + \frac{\sigma^2}{N}$$

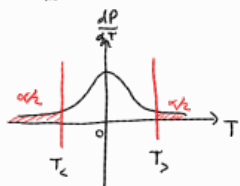
$$\rightarrow \langle s^2 \rangle = k \sum \left(\sigma^2 + \mu^2 - \left(\mu^2 + \frac{\sigma^2}{N} \right) \right) = k \sigma^2 N \left(1 - \frac{1}{N} \right) = k \sigma^2 (N-1) = \sigma^2$$

$$\rightarrow k = \frac{1}{N-1}$$

b) ocenjevanje intervalov zaupanja za pričak. vr. μ

• T-statistika

$$T = \frac{\bar{z} - \mu}{\frac{s}{\sqrt{N}}} \sim S(N-1) \text{ studentova porazd. z } N-1 \text{ prost. st.}$$



kritična vrednost za T-statistiko: T_c, T_s

α ... stopnja tveganja, $1-\alpha$... stopnja zaupanja

$$P(T > T_s) = \alpha/2$$

$$P(T < T_c) = \alpha/2$$

$$T_c = -T_s$$

$$P(|T| > T_s) = \alpha$$

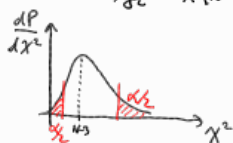
v tabelah: $P(|T| > T_s) = p$

	90%	95%	99%	...
$N-1$				
	T_s			

$$\rightarrow \mu = \bar{z} \pm T_s \frac{s}{\sqrt{N}} \text{ na stopnji tveganja } \alpha$$

• interval zaupanja za varianco σ^2

$$\chi^2 = (N-1) \frac{s^2}{\sigma^2} \sim \chi^2(N-1) \rightarrow \text{porazdelitev } \chi^2 \text{ z } N-1 \text{ prost. st.}$$



α ... stopnja tveganja

$$P(\chi^2 > \chi^2_s) = \frac{\alpha}{2}$$

$$P(\chi^2 < \chi^2_c) = \frac{\alpha}{2} \rightarrow P(\chi^2 > \chi^2_c) = 1 - \frac{\alpha}{2}$$

v tabelah: $P(\chi^2 > \chi^2_s) = p$

$$\rightarrow \sigma^2 = (N-1) \frac{s^2}{\chi^2_s} \text{ na stopnji tveganja } \alpha$$

obratno!

$$\langle \chi^2 \rangle = N-1$$

c) Testiranje hipotez

$$H_0: \mu = \mu_0, \{z_i\}_n$$

ob privzetku, da H_0 velja izračunamo T-statistiko $T = \frac{\bar{z} - \mu_0}{s/\sqrt{n}}$ pri stopnji tveganja α

• če $|T| > T_{\alpha/2}$: zavržemo H_0 na st. tr. α

• če $|T| \leq T_{\alpha/2}$: H_0 ne moremo zavrniti na st. tr. α

Primer: 2. kolokvij 2019

h [m]	a) $\alpha = 2\%$
1680	interval zaupanja za $h = ?$
1740	
1690	b) $1 - \alpha = 95\%$
1680	
1740	$\sigma_0 = 20$ m
1720	$H_0: \sigma^2 = \sigma_0^2 = 400$ m ²
	testiraj hipotezo

a)

$$N = 6$$

$$\bar{z} = \frac{1}{N} \sum z_i = 1700 \text{ m}$$

$$s^2 = \frac{1}{N-1} \sum (z_i - \bar{z})^2 = 840 \text{ m}^2$$

$$P(|T| > T_{\alpha/2}) = \alpha = 2\% \rightarrow T_{\alpha/2} = 3,365$$

$$\alpha = \bar{z} \pm T_{\alpha/2} \frac{s}{\sqrt{N}} = 1700 \text{ m} \pm 40 \text{ m}$$

b) $\alpha = 5\%$

$$\chi_0^2 = (N-1) \frac{s^2}{\sigma_0^2} = 10,5$$

$$P(\chi^2 > \chi_0^2) = \frac{\alpha}{2} = 2,5\% \rightarrow \chi_0^2 = 12,833$$

$$P(\chi^2 > \chi_0^2) = 1 - \frac{\alpha}{2} = 97,5\% \rightarrow \text{ne rabin, ker vemo da je maksimum porabljen pri } N=3$$

$$\Rightarrow \chi_0^2 = 10,5 > 3$$

$\chi_0^2 < \chi_0^2 < \chi_0^2 \Rightarrow$ hipoteze ne moremo zavrniti pri st. tr. 5%

Primer: 2. pismi izpit 2020

$$\alpha = 5\%, N = 10, \mu \in (10,6; 12,0)$$

a) $\bar{z}, s^2 = ?$

$T = ?$

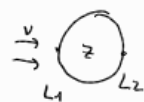
$$\alpha = \bar{z} \pm T_{\alpha/2} \frac{s}{\sqrt{N}}$$

$$a_1 = 12,0 = \bar{z} + T_{\alpha/2} \frac{s}{\sqrt{N}}$$

$$a_2 = 10,6 = \bar{z} - T_{\alpha/2} \frac{s}{\sqrt{N}}$$

$$T_{\alpha/2}(N=10, \alpha=5\%) = 2,228$$

Primer:



tok nevtrinov,
2 laboratorija

L1	L2
6,45	6,45
6,46	5,76
6,25	6,46
6,03	5,78
6,46	6,14
6,41	5,70

$$\alpha = 5\%$$

ali sta $L1$ in $L2$

merila isti tok nevtrinov?

$$H_0: \mu_1 = \mu_2$$

$$\text{predp.: } \sigma_1^2 = \sigma_2^2 = \sigma^2$$

T-statistika:

$$T_0 = \frac{(\bar{L}_1 - \bar{L}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{1}{N_1} + \frac{1}{N_2}}} \frac{\sqrt{N_1 + N_2 - 2}}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}} \sim S(N_1 + N_2 - 2)$$

$$T_0 = 2,254$$

$$T_{\alpha/2}(n=10, \alpha=5\%) = 2,228$$

$T_0 > T_{\alpha/2} \Rightarrow$ hipotezo zavrnemo na st. tr. α

Oblikovni testi

1) Pearsonov χ_p^2 -test

$$H_0: z_i \sim \frac{dP}{dz} \leftarrow \text{paži na normirano!}$$

• izmerki $\rightarrow$ histogram N_k ... št. izmerkov v k -tem razredu

$$\text{verjetnost, da } z_i \text{ pade v } k\text{-ti razred: } p_k = \int_{z_{k-1}}^{z_k} \frac{dP}{dz} dz$$

$\rightarrow$ pričakovano št. izmerkov v k -tem razredu: $N p_k$ pogoj, da test deluje: $N p_k \geq 5$

$$\chi_p^2 = \sum_{k=1}^g \frac{(N_k - N p_k)^2}{N p_k} \sim \chi^2(g)$$

$\rightarrow$ kritična vrednost: enostranski test $P(\chi^2 > \chi_0^2) = \alpha$

• $\chi_p^2 > \chi_0^2$: zavržemo H_0 na st. tr. α

primer: met kocke

$$\alpha = 1\%$$

ali je kocka pravična?

$$\rightarrow \frac{dP}{dz} = \text{konst.} \rightarrow p_k = \frac{1}{6}$$

$$N = 60$$

$$N p_k = 10$$

$$\chi_p^2 = \sum \frac{(N_k - N p_k)^2}{N p_k} = \frac{25 + 4 + 1 + 4 + 0 + 100}{10} = 13,4$$

$$\chi_0^2 = 15,086$$

$\Rightarrow$ ne moremo zavrniti hipoteze

Linearna metoda najin kvadratov

linearna v parametrih, odvisnosti so lahko nelinearne

meritve:

$$z(t) \rightarrow z_i = z(t_i) \rightarrow \vec{z} = \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix}$$

model:

$$z(t) = x_1 f_1(t) + x_2 f_2(t) + \dots + x_n f_n(t)$$

$$\vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \text{ išemo optimalno oceno za } \vec{x}$$

$$\vec{z} = H\vec{x} + \vec{r}$$

Ločenska (strukturirna) matrika

$$H = \begin{pmatrix} f_1(t_1) & f_2(t_1) & \dots & f_n(t_1) \\ f_1(t_2) & f_2(t_2) & \dots & f_n(t_2) \\ \vdots & \vdots & \ddots & \vdots \\ f_1(t_n) & f_2(t_n) & \dots & f_n(t_n) \end{pmatrix}$$

$$\hat{\vec{x}} = \vec{x} + K(\vec{z} - H\vec{x})$$

$$K = PH^T R^{-1}$$

$$P^{-1} = I + H^T R^{-1} H$$

$\vec{x}$ ne poznamo $\Rightarrow M \rightarrow \infty$

$$\rightarrow P^{-1} = H^T R^{-1} H, R = \sigma^2 I \text{ (privzetek)}$$

$$\rightarrow P = (H^T H)^{-1} \sigma^2$$

$$\rightarrow \hat{\vec{x}} = \vec{x} + PH^T R^{-1} (\vec{z} - H\vec{x}) = \frac{PH^T R^{-1} H \vec{x}}{P^{-1}}$$

$$= PH^T R^{-1} \vec{z}$$

$$\hat{\vec{x}} = (H^T H)^{-1} H^T \vec{z}$$

kvadratna forma:

$$2J = (\vec{z} - H\vec{x})^T R^{-1} (\vec{z} - H\vec{x})$$

$$\frac{\partial 2J}{\partial \vec{x}} = 0 \rightarrow \text{minimum kv. forme} \Rightarrow \hat{\vec{x}}$$

$$2J = \chi^2 = (\vec{z} - H\hat{\vec{x}})^T R^{-1} (\vec{z} - H\hat{\vec{x}}) =$$

$$= \frac{1}{\sigma^2} (\vec{z} - H\hat{\vec{x}})^T (\vec{z} - H\hat{\vec{x}}) \leftarrow \text{skal. pr}$$

$$= \sum_{i=1}^n \frac{(z_i - H\hat{\vec{x}}(t_i))^2}{\sigma^2} \sim \chi^2(n-m)$$

$$\rightarrow \langle 2J \rangle = n-m$$

$$\text{normirani } \chi^2: \chi^2_n = \frac{\chi^2}{\langle \chi^2 \rangle} = \frac{\chi^2}{n-m}$$

Primer:

model: $x_1 t$

$$\vec{x} = (x_1)$$

$$f_1(t) = t$$

$$H = \begin{pmatrix} t_1 \\ t_2 \\ \vdots \\ t_n \end{pmatrix}, \vec{z} = \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{pmatrix}$$

$$R = \sigma^2 I$$

$$P = (H^T H)^{-1} R$$

$$H^T H = \sum t_i^2$$

$$\rightarrow P = \frac{\sigma^2}{\sum t_i^2}$$

$$\hat{x} = \frac{\sum t_i z_i}{\sum t_i^2}$$

Primer:

model: $x_1 t + x_2$

$$\vec{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$H = \begin{pmatrix} t_1 & 1 \\ t_2 & 1 \\ \vdots & \vdots \\ t_n & 1 \end{pmatrix}$$

$$H^T H = \begin{pmatrix} \sum t_i^2 & \sum t_i \\ \sum t_i & n \end{pmatrix}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{\det} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$P = \frac{\sigma^2}{n \sum t_i^2 - (\sum t_i)^2} \begin{pmatrix} n & -\sum t_i \\ -\sum t_i & \sum t_i^2 \end{pmatrix}$$

$$\hat{\vec{x}} = \frac{1}{n \sum t_i^2 - (\sum t_i)^2} \begin{pmatrix} n \sum t_i z_i - \sum t_i \sum z_i \\ \sum t_i^2 \sum z_i - \sum t_i \sum t_i z_i \end{pmatrix} = \begin{pmatrix} \hat{x}_1 \\ \hat{x}_2 \end{pmatrix}$$

Primer:

v P_1 $\rightarrow$ parabola $H: P = \beta v^2 + P_0$

60 4,8

72 5,0

90 5,1

120 5,2

150 5,0

$$\alpha = 10\%$$

$$\sigma = 0,5 \frac{1}{100 \text{ km}}$$

$$v^2 = t$$

$$n=5$$

$$\sum t_i = \sum v_i^2 = 53784$$

$$\sum z_i = \sum P_i = 36,1$$

$$(\sum t_i)^2 = 289272 \cdot 10^4$$

$$\sum t_i z_i = \sum v_i^2 P_i = 466300$$

$$\sum t_i^2 = \sum v_i^4 = 81905 \cdot 10^4$$

$$\hat{\vec{x}} = \dots = \begin{pmatrix} 3,25 \cdot 10^{-4} \frac{1}{100 \text{ km}} (km/h)^{-2} \\ 3,73 \frac{1}{100 \text{ km}} \end{pmatrix} = \begin{pmatrix} \hat{\beta} \\ \hat{P}_0 \end{pmatrix}$$

$$\chi^2 = \frac{1}{\sigma^2} \sum (P_i - [\hat{\beta} v_i^2 + \hat{P}_0])^2 \approx 3,1$$

$$P(\chi^2 > \chi^2_{3, 10\%}) = 10\%$$

$$\rightarrow \chi^2_3 = 6,25$$

$\Rightarrow$ ne moremo zavrniti hipoteze

Primer:

nelinearen model: $I_i = I_0 e^{-t_i/\tau} \Rightarrow \ln \frac{I_i}{I_0} = -\frac{1}{\tau} t_i = x_1 t_i$

Primer:

Nekvadratnih meritev

($t_1=0, t_2=\Delta t, \dots, t_n=(n-1)\Delta t$)

model $k t + n$

$\alpha = 10\%, \chi^2_3 = 2,3512$

Doloci frekvenco vzorčenja $\nu = \frac{1}{\Delta t}$

$$\sigma_k = \sigma = 0,2 \text{ m}$$

$$\sigma_k = 0,02 \text{ s}$$

$$\text{povprečje si } \sum_{i=1}^m i = \frac{m(m+1)}{2}, \sum_{i=1}^m i^2 = \frac{m(m+1)(2m+1)}{6}$$

$$P = \begin{pmatrix} \sigma_k^2 & \sigma_{kn} \\ \sigma_{kn} & \sigma_n^2 \end{pmatrix}$$

$$H = \begin{pmatrix} t_1 & 1 \\ \vdots & \vdots \\ t_n & 1 \end{pmatrix}$$

$$P = \sigma^2 (H^T H)^{-1} = \frac{\sigma^2}{N \sum t_i^2 - (\sum t_i)^2} \begin{pmatrix} N & -\sum t_i \\ -\sum t_i & \sum t_i^2 \end{pmatrix}$$

$$\sigma_k^2 = \frac{N \sigma^2}{N \sum t_i^2 - (\sum t_i)^2}$$

$$\sum t_i = \sum (i-1) \Delta t = \Delta t \sum_{i=1}^{N-1} i = \Delta t \frac{N(N-1)}{2}$$

$$\sum t_i^2 = \sum_{i=1}^{N-1} ((i-1) \Delta t)^2 = \Delta t^2 \sum_{i=1}^{N-1} i^2 = \Delta t^2 \frac{N(N-1)(2N-1)}{6}$$

$$\sigma_k^2 = \frac{N \sigma^2}{\Delta t^2 \left(\frac{N^2(N-1)(2N-1)}{6} - \frac{N^2(N-1)^2}{4} \right)} =$$

$$= \frac{\sigma^2}{\Delta t^2} \cdot \frac{1}{N(N-1)} \cdot \frac{12}{2(2N-1) - 3(N-1)} =$$

$$= \frac{12 \sigma^2}{\Delta t^2} \frac{1}{N(N^2-1)}$$

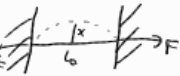
$$\Delta t = \frac{\sigma_k}{\sigma} \sqrt{\frac{12}{N(N^2-1)}} \rightarrow \nu = \frac{1}{\Delta t} \approx 3 \text{ Hz}$$

$$\alpha = 10\%, \chi^2_3 = 2,3512 \Rightarrow N-M=20$$

$$M=2 \Rightarrow N=22$$

Ocenjevanje natančnosti

Primer:



$x \ll l_0$
oceni natančnost $\left(\frac{\Delta v}{v}\right)$
zvečenja strune (t. nih. učinj)

$$c = \lambda v$$

$$v = \frac{c}{2l_0}$$

$$c = \sqrt{\frac{F}{\mu}}, \mu = \frac{m}{l_0}$$

$$\rightarrow v = \frac{1}{2l_0} \sqrt{\frac{F}{\mu}} = f(F, \mu)$$

$$v + \Delta v = \frac{1}{2l_0} \sqrt{\frac{F + \Delta F}{\mu + \Delta \mu}} = \frac{1}{2l_0} \sqrt{\frac{F}{\mu}} \sqrt{\frac{1 + \frac{\Delta F}{F}}{1 + \frac{\Delta \mu}{\mu}}} =$$

$$= v \left(1 + \frac{1}{2} \frac{\Delta F}{F}\right) \left(1 - \frac{1}{2} \frac{\Delta \mu}{\mu}\right) =$$

$$= v \left(1 + \frac{1}{2} \frac{\Delta F}{F} - \frac{1}{2} \frac{\Delta \mu}{\mu} - \frac{1}{8} \frac{\Delta F}{F} \frac{\Delta \mu}{\mu}\right)$$

$$\frac{\Delta v}{v} = \frac{1}{2} \frac{\Delta F}{F} - \frac{1}{2} \frac{\Delta \mu}{\mu}$$

$$\mu = \frac{m}{l_0}$$

$$\mu + \Delta \mu = \frac{m}{l_0 + \Delta l} = \frac{m}{l_0} (1 - \frac{\Delta l}{l_0}) = \mu \left(1 - \frac{\Delta l}{l_0}\right)$$

$$\frac{\Delta \mu}{\mu} = - \frac{\Delta l}{l_0}$$

$$\frac{\Delta v}{v} = \frac{1}{2} \left(\frac{\Delta F}{F} + \frac{\Delta l}{l_0}\right) = \frac{1}{2} \frac{\Delta l}{l_0} \left(1 + \frac{ES}{F}\right)$$

Hooke:

$$\frac{\Delta F}{S} = E \frac{\Delta l}{l_0}$$

$$\frac{l_0 + \Delta l}{2} x \rightarrow x^2 + \frac{l_0^2}{4} = \frac{(l_0 + \Delta l)^2}{4}$$

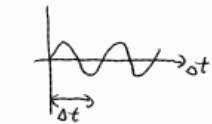
$$\frac{\Delta l}{l_0} = \frac{\sqrt{4x^2 + l_0^2} - l_0}{l_0} \approx 2 \left(\frac{x}{l_0}\right)^2$$

$$\rightarrow \frac{\Delta v}{v} = \left(\frac{x}{l_0}\right)^2 \left(1 + \frac{ES}{F}\right)$$

Ocenjevanje natančnosti s principom nedoločnosti

svot 31

↳ širina okna
↳ pasovna širina



$$\Rightarrow \Delta v \approx \frac{1}{\Delta t}$$

Primer: koliko časa moramo poslušati ton
z neznano frekvenco (540 Hz), da ga
določimo na pod tem natančnost?

$$v_1 = 266$$

$$\text{podatki: } \frac{v_{i+1}}{v_i} = x$$

$$\Delta v = v_1 - v_0 = x v_0 - v_0 = v_0 (x - 1)$$

$$\Delta t = \frac{1}{v_0 (x - 1)} = \frac{1}{v_0 (2^{1/n} - 1)} = \frac{1}{266 \text{ Hz}} = \underline{\underline{40 \text{ ms}}}$$

$$\frac{v_1}{v_0} = x^2 \Rightarrow \frac{v_1}{v_0} = x^4$$

$$\frac{v_2}{v_0} = x^{12} = 2$$

$$x = 2^{1/12}$$